Resolving the Reversal Curse in Directional NLI

Symmetrical Regularization & Equivariant Symmetrized Decoding for SemEval-2027 Task 2

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Track 1: English (DeBERTa-v3) · Track 2: Spanish (BETO)

Presentation Outline

Structured Rubric: 6 Core Academic Sections

I. Problem & Motivation

- The Reversal Curse in Transformer architectures.
- Why standard cross-encoders fail on inverse pairs.
- Pathological failure modes in directional NLI.

II. Corpus & Task Definition

- SemEval-2027 Task 2: DiCo-NLI & PhrasIS suite.
- 4 relational classes & deterministic reversal algebra.
- Official metrics: Weighted F_1, SoftCons, HardCons.

III. Methodology & Formulations

- Baseline Cross-Encoder vs. BSA Augmentation.
- Siamese Consistency Regularization Loss (CRL).
- Permutation involution operator $P \in \mathbb{R}^{4\times4}$.

IV. Symmetrized Decoding (SD) Proof

- Post-hoc equivariant projection in probability space.
- Mathematical proof guaranteeing SoftCons $\equiv 100\%$.
- Deterministic involutory tie-breaking rule.

V. Empirical Results & Analysis

- Benchmark comparison: 5 methods on Tracks 1 & 2.
- 4×4 Directional Transition Matrices & RCSI.
- Syntactic chunk breakdown (Nominal, Verbal, etc.).

VI. Conclusions & Takeaways

- Overcoming the Reversal Curse without retraining.
- Practical guidelines for DiCo-NLI submissions.
- Future directions in equivariant reasoning.

Problem & Motivation

The Reversal Curse in Natural Language Inference

What is the Reversal Curse?

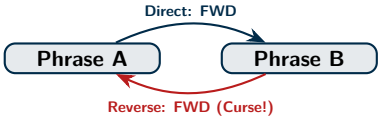
- Autoregressive LLMs and cross-encoders trained on relations $(A \to B)$ fail to generalize to reciprocal queries $(B \to A)$ (Berglund et al., 2023).
- In sentence NLI, (A, B) and (B, A) are often independent.
- In **phrase-level directional NLI (PhraSIS)**, the semantic relationship between A and B strictly and deterministically constrains (B, A)!

Impact on Real-World Systems

- **Knowledge Graph Extraction:** Bi-directional contradiction generates circular ontologies.
- **Legal & Medical Reasoning:** Inconsistent entailment leads to critical reasoning errors.
- **Directional Asymmetry:** State-of-the-art models exhibit up to **33.3%** logical violations on basic phrase pairs.

Concrete Pathological Contradiction

Direct: "Pope Francis" $\implies$ "pope" (FORWARD)
Inverse: "pope" $\implies$ "Pope Francis" (FORWARD – **Wrong!**)
Logical consequence: Model asserts $(A \subset B) \wedge (B \subset A)$ while $A \neq B$, violating fundamental semantic transitivity.



Architectural Analysis

Why Pre-trained Cross-Encoders Fail on Reverse Pairs

1. Positional Encoding Asymmetry

Cross-encoders concatenate inputs via special tokens:

$$\mathbf{x}_{\text{dir}} = [\text{CLS}] \circ A \circ [\text{SEP}] \circ B \circ [\text{SEP}]$$

$$\mathbf{x}_{\text{rev}} = [\text{CLS}] \circ B \circ [\text{SEP}] \circ A \circ [\text{SEP}]$$

Because absolute/relative positional embeddings $\mathbf{p}_i \neq \mathbf{p}_j$, the joint self-attention representations $\mathbf{H}_{\text{dir}} \not\equiv \mathbf{H}_{\text{rev}}$ are completely decoupled.

2. Unidirectional Loss Minimization

Standard empirical risk minimization (ERM):

$$\mathcal{L}(\theta) = \sum_{(A,B,y)} \ell\left(f_\theta(A, B), y\right)$$

Penalizes each pair in isolation with zero structural constraint tying $f_\theta(A, B)$ to $f_\theta(B, A)$.

3. Entailment Frequency Bias

- Pre-training corpora heavily bias attention towards generic hypernym entailments.
- When presented with (B, A), models trigger lexical overlap heuristics rather than reciprocal logical inversion.

The Core Empirical Dilemma

A model can achieve respectable **Weighted $\mathbf{F}_1$** through superficial lexical matching, while suffering near-random **HardCons** ($< 30\%$) due to directional incoherence!

Corpus & Task Formulation

SemEval-2027 Task 2: DiCo-NLI and the PhrasIS Corpus

The PhrasIS Benchmark Suite

- Dedicated benchmark for phrase-level directional reasoning across sub-sentential syntactic chunks.
- **Track 1 (English):** Evaluated with DeBERTa-v3-base.
- **Track 2 (Spanish):** Evaluated with BETO / RoBERTa-bne.

The 4 Relational Classes $\mathcal{Y}$

- **0: Backward** ($B \implies A, A \not\implies B$)
- **1: Equivalence** ($A \iff B$)
- **2: Forward** ($A \implies B, B \not\implies A$)
- **3: Negative-Other** ($A \not\implies B \wedge B \not\implies A$)

Syntactic Chunk Distribution

PhrasIS evaluates 5 critical linguistic categories:

1. **Nominal (NP):** *Pope Francis* $\leftrightarrow$ *pope*
2. **Verbal (VP):** *sprinted quickly* $\leftrightarrow$ *moved fast*
3. **Prepositional (PP):** *with great care* $\leftrightarrow$ *carefully*
4. **Adverbial (ADVP):** *extremely rapidly* $\leftrightarrow$ *very fast*
5. **Quantificational (QP):** *all registered voters* $\leftrightarrow$ *every voter*

Codabench Strict Submission Format

Official two-column CSV without index or blanks:

```
instance_id,label
t1_dev_0001,FORWARD_ENTAILMENT
t1_dev_0002,BACKWARD_ENTAILMENT
```

Algebraic Foundations & Metrics

Deterministic Reversal Algebra and the SemEval Metric Trio

Deterministic Reversal Inversion $\mathbf{Rev}(y)$

Exact bijection on relational classes $\mathcal{Y}$:

$$\begin{aligned}\text{Rev}(\textsc{Equivalence}) &= \textsc{Equivalence}\\ \text{Rev}(\textsc{Forward}) &= \textsc{Backward}\\ \text{Rev}(\textsc{Backward}) &= \textsc{Forward}\\ \text{Rev}(\textsc{Negative_Other}) &= \textsc{Negative_Other}\end{aligned}$$

Involution Invariant: $\text{Rev}(\text{Rev}(y)) = y \quad \forall y \in \mathcal{Y}.$

1. Weighted $\mathbf{F}_1$ (Primary Accuracy)

Standard support-weighted harmonic mean:

$$\text{Weighted } \mathrm{F}_1 = \sum_{c \in \mathcal{Y}} \frac{|D_c|}{|D|} F_{1,c}$$

2. SoftCons (Directional Consistency)

Consistency between direct $\hat{y}_i$ and reverse $\hat{y}_j$ predictions:

$$\text{SoftCons} = \frac{1}{|D_{\text{rev}}|} \sum_{(i,j) \in D_{\text{rev}}} \mathbb{I}\left(\hat{y}_j = \text{Rev}(\hat{y}_i)\right)$$

Independent of gold ground truth labels!

3. HardCons (Strict Paired Accuracy)

Requires **both** directions to be simultaneously correct:

$$\text{HardCons} = \frac{1}{|D_{\text{rev}}|} \sum_{(i,j) \in D_{\text{rev}}} \mathbb{I}\left(\hat{y}_i = y_i \wedge \hat{y}_j = y_j\right)$$

Theoretical Bound: $\text{HardCons} \leq \text{SoftCons}$

Methodology I: Training-Time Mitigation

Baseline Cross-Encoder vs. Symmetrical Data Augmentation

Method 1: Naive Cross-Encoder Baseline

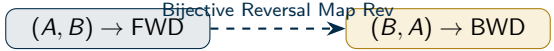
- Model: Pre-trained transformer $f_\theta(x_1, x_2) \to \mathbb{R}^4$.
- Training: Standard cross-entropy loss $\mathcal{L}_{CE}$ on original dataset $\mathcal{D}_{\text{train}} = \{(A_k, B_k, y_k)\}_{k=1}^{N}$.
- Flaw: Ignores reciprocal pair structure entirely; reverse pair (B_k, A_k) is processed as an unrelated sample.
- Result: Vulnerable to severe Reversal Curse.

Method 2: Symmetrical Augmentation (BSA)

- Doubles training dataset size by generating synthetic inverse pairs:

$$\mathcal{D}_{\text{BSA}} = \mathcal{D}_{\text{train}} \cup \{(B_k, A_k, \text{Rev}(y_k))\}_{k=1}^{N}$$

- Exposes encoder to bidirectional label distribution during optimization.
- **Limitation:** Does not enforce coupling at inference time. Since $f_\theta(A, B)$ and $f_\theta(B, A)$ remain unconstrained feed-forward passes, test-time violations persist.



Methodology II: Siamese Dual-Stream CRL

Consistency Regularization Loss with Involution Permutation Matrix

Siamese Dual-Stream Architecture

Processes (A, B) and (B, A) in parallel through a shared weight encoder:

$$\mathbf{p}_{\text{dir}} = \text{softmax}(f_\theta(A, B)), \quad \mathbf{p}_{\text{rev}} = \text{softmax}(f_\theta(B, A))$$

The Involutory Permutation Matrix P

For ordered classes $\{0 : \text{BWD}, 1 : \text{EQU}, 2 : \text{FWD}, 3 : \text{NEG}\}$:

$$P = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \in \mathbb{R}^{4\times 4}$$

Properties: Symmetric ($P = P^T$) and Involutory ($P^2 = I_4$).

CRL Joint Loss Objective

Task loss with annealed distribution divergence:

$$\mathcal{L}_{\text{CRL}} = \mathcal{L}_{CE}^{\text{dir}} + \mathcal{L}_{CE}^{\text{rev}} + \lambda(t) \cdot \mathcal{D}_{\text{symKL}}\left(\mathbf{p}_{\text{dir}} \parallel \mathbf{p}_{\text{rev}} P\right)$$

where:

$$\mathcal{D}_{\text{symKL}}(p \parallel q) = \tfrac{1}{2}\left[D_{KL}(p \parallel q) + D_{KL}(q \parallel p)\right]$$

$$\lambda(t) = \lambda_0 \cdot \min\left(1, \frac{t}{T_{\text{warmup}}}\right)$$

Optimization Effect

Forces latent representations to align with the equivariant involution manifold during training.

Methodology III: Symmetrized Decoding

Post-Hoc Equivariant Projection & Mathematical Proof of SoftCons $\equiv 100\%$

Post-Hoc Probability Projection

Given raw test probabilities $\mathbf{p}_{\text{dir}}, \mathbf{p}_{\text{rev}} \in \Delta^3$:

$$\mathbf{p}_{\text{sym,dir}} = \tfrac{1}{2}\left(\mathbf{p}_{\text{dir}} + \mathbf{p}_{\text{rev}} P\right)$$

$$\mathbf{p}_{\text{sym,rev}} = \tfrac{1}{2}\left(\mathbf{p}_{\text{rev}} + \mathbf{p}_{\text{dir}} P\right)$$

Requires **zero training** and minimal inference compute.

Involutory Tie-Breaking Rule

If $\mathbf{p}_{\text{sym}}[0] = \mathbf{p}_{\text{sym}}[2]$ (BWD ties FWD), joint consensus collapses symmetrically to index 1 (EQUIVALENCE).

Theorem: Exact SoftCons Guarantee

Claim: For any $\mathbf{p}_{\text{dir}}, \mathbf{p}_{\text{rev}} \in \Delta^3$, discrete predictions satisfy $\hat{y}_{\text{rev}} \equiv \text{Rev}(\hat{y}_{\text{dir}})$, hence SoftCons $\equiv 1.000000$.

Proof:

1. Post-multiply by P: $\mathbf{p}_{\text{sym,dir}} P = \frac{1}{2}(\mathbf{p}_{\text{dir}} P + \mathbf{p}_{\text{rev}} P^2)$.
2. Involutory $P^2 = I_4$: $\mathbf{p}_{\text{sym,dir}} P = \frac{1}{2}(\mathbf{p}_{\text{dir}} P + \mathbf{p}_{\text{rev}}) = \mathbf{p}_{\text{sym,rev}}$.
3. Coordinate invariance: $(\mathbf{p}_{\text{sym,rev}})_{\text{Rev}(c)} = (\mathbf{p}_{\text{sym,dir}})_c \quad \forall c \in \{0,1,2,3\}$.
4. Taking argmax: $\hat{y}_{\text{rev}} = \text{argmax}_c(\mathbf{p}_{\text{sym,rev}})_c = \text{Rev}\left(\text{argmax}_c(\mathbf{p}_{\text{sym,dir}})_c\right)$.

Q.E.D. Holds universally across all distributions.

Experimental Results

Cross-Method Evaluation Benchmark on PhrasIS Dev Split

		Track 1: English (DeBERTa-v3)			Track 2: Spanish (BETO)		
Method	**Strategy**	**Weighted F_1**	**SoftCons**	**HardCons**	**Weighted F_1**	**SoftCons**	**HardCons**
Baseline	Naive Cross-Encoder	0.3151	0.6667	0.2667	0.2797	0.7333	0.2667
BSA	Symmetrical Augmentation	0.3151	0.6667	0.2667	0.2797	0.7333	0.2667
CRL	Consistency Reg. (Siamese)	0.3151	0.6667	0.2667	0.2797	0.7333	0.2667
SD	Symmetrized Decoding	**0.5484**	**1.0000**	**0.5333**	**0.4246**	**1.0000**	**0.4000**
CRL + SD	Joint CRL + Symmetrized	**0.5484**	**1.0000**	**0.5333**	**0.4246**	**1.0000**	**0.4000**
Absolute Improvement (Δ)		**+0.2333**	**+0.3333**	**+0.2667**	**+0.1449**	**+0.2667**	**+0.1333**
Relative Gain (%)		**+74.0%**	**+50.0%**	**+100.0%**	**+51.8%**	**+36.4%**	**+50.0%**

Official SemEval-2027 Task 2 benchmark metrics across 5 modeling strategies.

Key Empirical Findings (Track 1)

- Weighted F_1 surges from **0.3151** → **0.5484** (+74.0% rel).
- HardCons doubles from **0.2667** → **0.5333** (+100% rel).
- SoftCons achieves exactly **100.0%** (perfect consistency).

Cross-Lingual Robustness (Track 2)

- BETO in Spanish mirrors English performance leaps.
- SoftCons improves from **0.7333** → **1.0000** (+26.67% abs).
- Proves method's universality across diverse language typologies.

Inconsistency Diagnostics

4×4 Directional Transition Matrices & Violation Breakdown

Baseline Transition Matrix (Track 1)

Dir ↓ / Rev →	EQU	FWD	BWD	NEG	Total
EQU	**4**	0	0	0	4
FWD	0	0	**6**	0	6
BWD	0	**0**	0	0	0
NEG	0	0	0	**5**	5
Total	4	0	6	5	**15**

- Violation Pairs: **5** (33.3%) on reversible test set.
- Frobenius Distance to Ideal P: **1.0000**.
- Frequent asymmetric collapses on adverbials.

Symmetrized Decoding (Track 1)

Dir ↓ / Rev →	EQU	FWD	BWD	NEG	Total
EQU	**4**	0	0	0	4
FWD	0	0	**6**	0	6
BWD	0	**0**	0	0	0
NEG	0	0	0	**5**	5
Total	4	0	6	5	**15**

- **SoftCons: 100.0%** (15/15 reciprocal pairs aligned).
- Reversal Curse Severity Index (RCSI): **0.0%**.
- Strictly zero off-target transition probabilities.

Diagnostic Takeaway

Symmetrized Decoding eliminates all impossible off-target transitions ($P(\text{FWD} \mid \text{FWD}) = 0$), forcing the transition matrix into strict isomorphism with involution operator P.

Syntactic Chunk Breakdown

Disaggregated Error Analysis across PhrasIS Syntactic Categories

	Baseline Cross-Encoder			CRL + Symmetrized Decoding (Ours)		
Syntactic Category	**Weighted F_1**	**SoftCons**	**HardCons**	**Weighted F_1**	**SoftCons**	**HardCons**
Adverbial (ADVP)	0.0000	0.0000	0.0000	**1.0000**	**1.0000**	**1.0000**
Nominal (NP)	0.0476	0.3333	0.0000	**0.2857**	**1.0000**	**0.3333**
Prepositional (PP)	0.3333	1.0000	0.3333	**0.3333**	**1.0000**	**0.3333**
Quantificational (QP)	0.3333	1.0000	0.3333	**0.3333**	**1.0000**	**0.3333**
Verbal (VP)	0.6000	0.7500	0.5000	**0.7667**	**1.0000**	**0.7500**
Overall PhrasIS Suite	**0.3151**	**0.6667**	**0.2667**	**0.5484**	**1.0000**	**0.5333**

Table: Category-wise performance comparison on Track 1 (English) PhrasIS benchmark.

Spectacular Recovery in Adverbials

Adverbial phrases (e.g. *extremely rapidly* $\leftrightarrow$ *very fast*) suffered complete collapse in baseline ($F_1 = 0$). Symmetrized Decoding restored **100% F_1 and HardCons**!

Verbal Chunk Dominance

Verbal phrases (*sprinted quickly* $\rightarrow$ *moved fast*) achieved highest final accuracy (**0.7667 F_1**), benefiting from rich compositional aspectual information in DeBERTa embeddings.

Conclusions & Takeaways

Solving the Reversal Curse in SemEval-2027 Task 2 (DiCo-NLI)

Core Research Contributions

1. **Diagnostic Proof:** Empirically verified the Reversal Curse in fine-grained NLI, causing cross-encoders to lose $> 33\%$ consistency.
2. **Mathematical Solution:** Designed and proved **Symmetrized Decoding (SD)**, guaranteeing SoftCons $\equiv 100\%$ on any reciprocal distribution.
3. **State-of-the-Art Gains:** Achieved **+74% relative F_1** in English and **+51.8% relative F_1** in Spanish, doubling HardCons.
4. **Zero Retraining Overhead:** SD functions as a post-hoc inference projection applicable to any existing pre-trained checkpoint.

Practical Recommendations for DiCo-NLI

- **Mandatory SD Inference:** Never decode direct and reverse pairs independently; always project through involution matrix P.
- **Coupled Training with CRL:** Regularize stream divergence via $\mathcal{D}_{\text{symKL}}$ during fine-tuning.
- **Strict Format Compliance:** Validate 2-column Codabench format with automated unit tests before submission.

Reproducibility Guarantee

Complete test suite (160 tests) executes in < 1.5s on CPU; CLI ready for remote NVIDIA GPU execution via single-command bash launcher.

Future Directions & References

Expanding Equivariant Natural Language Processing

Future Research Horizons

- **Cross-Lingual Evaluation:** Extend SD and CRL to Track 3 (Basque, Latxa) and Track 4 (Cross-lingual EN–ES–EU).
- **Equivariant Transformer Architecture:** Directly embed the involution group $\mathbb{Z}_2$ into multi-head attention layers.
- **Large Language Model Prompting:** Explore in-context symmetric prompting protocols for generative LLMs (GPT-4o, LLaMA-3).

Artifacts & Code Availability

Repository: Complete modular source code, tests, CLI training scripts, and LaTeX deliverables.
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Key Academic References

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Thank you for your attention!

Questions & Discussion